Barrier Fates | Botanische Tuinen | Universiteit Utrecht

Honest conversations with scientists, Part 3

How can science help people embrace new technologies?

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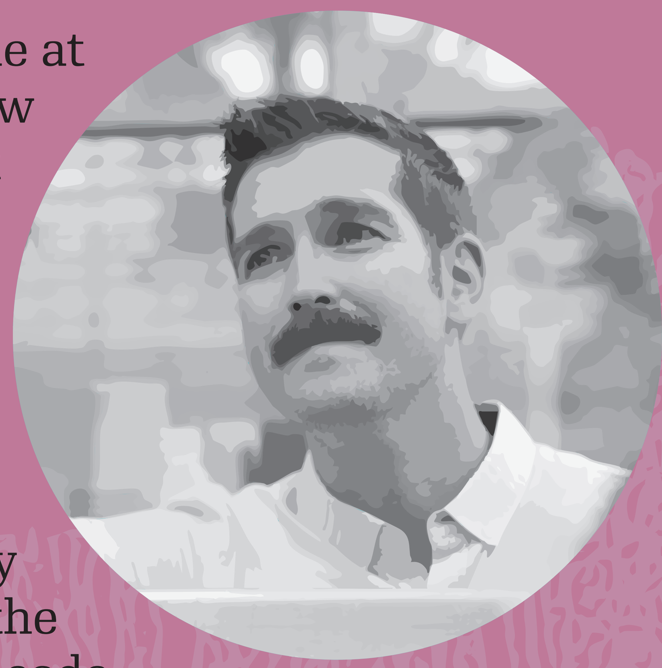

My research topic centers on the decisions made at the cellular level during plant development. How can one cell be so different from the one next to it when both have identical genetic code? Why is one cell red while the one next to it is white? What controls the pattern? Why does one cell become a leaf, while the other becomes a petal? How does this change in different plants and organs? These are the questions that will help us understand how biological systems themselves form the diverse array of shapes, colors, and sizes. Once we understand the programming, we will add our own programs to the code. We work with Chrysanthemum flowers, orchids, tomatoes and petunia flowers.

Want to see more flowers?

ON THE IMPACT

It's always fun to discuss my work with people from different backgrounds. Sharing what's possible from a scientific perspective and in turn hearing people's perspective is part of science's role in society. Ideally, more non-scientists could connect research to tangible improvements in people's lives.

For me, the biggest impact you can have is actually working on plants. Plants are everywhere. We depend on them for everything. You have to eat, and breathe oxygen.

Fun fact: if you've ever eaten a papaya, you've likely consumed a GMO. A virus once threathened to wipe papaya out. A genetic modification was the only way to save the plant.

ON THE CHALLENGES

People seem to embrace science and technology differently, depending on what it is applied to. For example, new technologies in face creams are widely accepted, but new technologies in tomato production are not. This is especially true for food, which is deeply tied to culture. Even well-intentioned changes can feel negative, because of that strong cultural connection.

The GMO debate illustrates this well. GMOs are simply a tool, neither inherently good nor bad. Debating them in absolute terms to me is like arguing whether we should use hammers at all; it is silly and misses the point.

ON PREVIOUS EXPERIENCES

Throughout my career, I've had the chance to talk to many types of audiences, and it has been a blast. During my PhD, I taught biology from elementary school to high school. It was amazing to see the kids connecting dots based on what I was teaching and getting excited themselves.

In these three images, there are three petunia flowers with different speck patterns. The change in patterns very visually shows the change in programming in the genetic code that I mentioned in the topmost section of this poster.

ON THE FUTURE

I think there are a few things we can do as scientists for science communication:

Improve our branding to help people distinguish what is and what isn't truly concerning, and push back against harmful corporate practices to protect farmers who would use GMOs.

Diversify academia by including communicators and educators besides traditional roles (PhDs, postdocs, PIs) to broaden impact.

Make better use of social media, for example by creating accounts for research groups to share their work and reach wider audiences.

Encourage open discussions about taboo topics, with a focus on mutual understanding.

Read this poster in Dutch

What do you think?